CESSDA Research Infrastructure

Technical Framework

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What is the UK Data Service?

The UK Data Service provides unified access to the UK's largest collection of social, economic and population data.

We provide:

- Access to high-quality local, regional, national and international social and economic data
- Support for policy-relevant research in the higher education, public and commercial sectors
- Guidance and training for the development of skills in data use
- The development of best practice data preservation and sharing standards
- The sharing of expertise with international data providers to remove barriers to accessing data

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Who Delivers the UK Data Service?

The UK Data Service is delivered by staff with specialist expertise in research data based at higher education institutions across the UK including:

- UK Data Archive, University of Essex
- Jisc
- The Cathie Marsh Institute for Social Research (CMIST), University of Manchester
- School of Geography, University of Leeds
- Geography and the Environment, University of Southampton
- EDINA, University of Edinburgh
- Department of Information Studies, University College London
- Centre for Advanced Spatial Analysis, University College London

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Structure

Introduction to CESSDA

Technical Framework (Phase 1)

What’s next?

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What is CESSDA?

Consortium of European Social Science Data Archives (http://cessda.net)

Currently 15 members, 1 observer

Aim: Provide a pan-European Research Infrastructure for the Social Sciences

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CESSDA Objectives

- coordination of the network of European data service providers and promotion of the results of social sciences
- facilitating researcher access to important resources of relevance to the European social science research agenda regardless of the location of either researcher or data
- work continuously to include further data sources from Europe and beyond, into the infrastructure

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CESSDA Objectives

- provide training within CESSDA and beyond on best practices surrounding operational processes and data management
- promote and facilitate wider participation in CESSDA
- the development and coordination of standards, protocols and professional best practice pertaining to the preservation and dissemination of data and associated digital objects

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CESSDA Current Activity

Horizon 2020 Projects

- CESSDA Strengthening and Widening (SaW)
- Synergies for Europe's Research Infrastructures in the Social Sciences (SERISS)
- Big Data Europe (BDE)

Work plan Projects

- Expert Seminars
- Metadata Harvester
- Metadata Management
- Data Access Policy
- Technical Framework

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Technical Infrastructure Required

In past, relied on members (‘Service Providers’) to develop and host standalone products

Now a company under Norwegian Law and in process of becoming an ERIC (European Research Infrastructure Consortium)

=> Greenfield site for technical development

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Protection of Assets is Vital

Ensure CESSDA has access to

- source code
- configuration files
- technical documentation

that underpin its products and services

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CESSDA's common interoperability characteristics

1. **Loosely coupled but coordinated** - enable Service Providers to retain independence, yet fully interact in an integrated service
2. **Sustainable** - enable medium and long term investment and business change decisions to be made.
3. **Extensible** - enable additional services to be built on or around it, including adapting to changing functional requirements over time.
4. **Maintainable** - enable components to be updated when IT specifications change.
5. **Standards based** - enable the coordinated and planned change to all the coupled, but coordinated, services.

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What is the Technical Framework?

A guide for the development of the various (software) products and services that form part of the CESSDA Research Infrastructure

- promote good practice for software development
- meet common interoperability characteristics

Infrastructure for Development, Staging and Production

- harmonise development tool chain for SPs

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Partners and Activities

Partners DANS, GESIS, NSD, SND, UKDS (lead)

All partners

- Develop and publish Technical Architecture

UKDS and NSD

- Specify, install and configure development infrastructure

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Infrastructure for Development (to date)

- Email Identity c/w Avatar
- Collaborative document creation/management
- Source code management
- Issue Tracker
- Wiki
- File storage
- Chatroom
- Continuous Integration Server
- System Availability Monitor
- Micro-service registry

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Quality Control

Software Maturity Levels

- ensure quality of the research infrastructure is maintained
- guidance on minimum, expected and excellent standards
- originally based on NASA's RRLs
- revised in light of 'Capability Development Model' from CESSDA SaW project

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Common Interoperability Characteristics

Achieved by:

- REST APIs c/w API design standards
- Architectural standards
- Common development environment
- Adoption of 12 Factor App principles
  - methodology for building Software-as-a-Service apps

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1. Loosely coupled but coordinated

Adopt (micro) services architecture based on RESTful web service APIs

- provides a mechanism for reusing and combining software artefacts

See also 12 factor app, number 7 (Port binding - Export services via port binding)

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2. Sustainable

The provision of common standards

- Technical Architecture document

Common development and test environment

- via the technical infrastructure

Deployment environment

- via extensions to the technical infrastructure

Central source-code repository

See also 12 factor app, number 1 (Codebase - One codebase tracked in revision control, many deploys)

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3. Extensible

Service API is key

- Integration point for new services
- Combination point for building new features

Version and support two versions simultaneously

- Allows services to evolve, without breaking contract provided to consumers.

See also 12 factor app, number 8 (Concurrency - Scale out via the process model)

See also 12 factor app, number 9 (Disposability - Maximize robustness with fast startup and graceful shutdown)

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4. Maintainable

Again, service API is key

- implementation of a service can be changed as required, to take advantage of developments in software technology
- location of services can be changed as required, to take advantage of developments in hardware technology.

See also 12 factor app, number 2 (Dependencies - Explicitly declare and isolate dependencies)

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5. Standards Based

- Provision of common architectural standards (via Technical Architecture)
- A consistent (in both the calling and return structures and formats) and versioned API

See also 12 factor app, number 4 (Backing services - Treat backing services as attached resources)

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Software Acceptance Criteria

- Documentation
- Intellectual property issues
- Extensibility
- Modularity
- Packaging
- Portability
- Standards compliance
- Support
- Verification and testing
- Internationalisation and Localization

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Software Maturity Levels

1 - Initial usability; software use is not recommended.

2 - Use is feasible; the software can be used by skilled personnel but with considerable effort, cost and risk.

3 - Use is possible by most users; with some effort, cost, and risk. A risk assessment should be made before use.

4 - Software is usable; with little effort, cost, and risk.

5 - Demonstrable usability; there is clear evidence that the software is widely used by many users.

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Software Maturity Levels

Intellectual Property (IPR)	Extensibility (EXT)	Modularity (MAR)	Packaging (PAC)	Portability (PAR)	Standards Compliance (CMP)	Support (SUP)	Verification and Testing (VAT)	Security (SAD)	Internationalisation and Localisation (IOR2)
Software developers have been identified and their responsibilities have been determined. Relevant notices of developer organisation(s) (or developers) have been reviewed for applicability to intellectual property rights. There maybe evidence of a draft intellectual property rights agreement that would result from cooperative activities with other developer organisations (or developers). Rights are not specified internally in the source code or externally in documentation, use rights or limitations have not been specified.	The software was not designed with extensibility in mind, so there is either no ability to extend or modify program behavior, or it is very difficult to do, even for usages similar to those of the software. Core design execution parameters cannot be changed. There is, so, or limited, availability of the source code; the logical flow of code may be hard to follow, with little to no cohesion.	There is evidence that the source code was written with no design or consideration for organising the code in terms of functionality for modularity or use. It may have been a demonstrator or pilot project.	Only source code or executable available, i.e. no packaging. There is no, or incomplete, installation documentation and no auto-build/installation facility is available. Even an experienced user may have difficulties installing the software.	The software as a whole is not portable. However, if there is some source code provided with sufficient internal, external and environment documentation is available then elements may be portable. Executable binaries are provided for a specific platform, but there is no useful information on porting. Porting as a whole is not feasible (e.g. due to licensing) or prohibitively expensive.	The software and software development process complies, at least in part, with locally defined standards and best practices. The standards may be internally or externally developed, but are implemented with modifications to meet local conditions. There may be little or no documented evidence of standards used, but it maybe possible to infer this from the software maintenance of functionality and interfaces.	There is known contact information available (to the developer organisation(s)) and there is an willingness to provide minimal, occasional support without guarantees. It may not be possible for an end-user to use the software without some support.	Software application form tested and unit testing performed.	Security was addressed in the development phase up to and including design.	Internationalisation and Localisation not addressed
Developer organisation(s) (or developers) have an agreement that addresses the potential conflicts of the proposed intellectual property rights and responsibilities for development. A limited rights statement has been drafted, and applied inconsistently in documentation and source code. Developer organisation(s) (or developers) may be contacted by regulator rights for use.	There is some consideration to extensibility, but that may only need for a limited number of use cases. Through use of methods such as object-oriented design or other tools which provide logical cohesion. Some extensibility is possible through configuration changes; isolation of configuration parameters and constants in clearly identified sections of source code; and/or limited opportunity for software modification.	There is no distinction between generic and solution-specific functionality. The source code is organised into a primary system that provides general functionality and one or two subsystems that each provide multiple, unrelated, functions; code within each module contains many independent logical parts. The architecture is closed with only a few internal functions accessible by external programs through the primary system.	Software includes auto-build features, but is only available for one operating system. Detailed installation instructions are available and building for other operating systems is possible for an experienced user.	The complete source code is available, without external dependencies that are not portable, but the software cannot be ported without significant changes to the software or the target context. Documentation on porting is insufficient, and should be taken into consideration when assessing the effort required. Cost-benefit and risk analysis, compared to rewriting for the new context and use cases, should be undertaken before porting is considered, and only initiated if the net benefits of using the software slightly outweigh the cost of developing new software. Porting will nevertheless require significant effort.	The software and software development process endeavour to comply with recognised standards or widely used best practices, but without verification or testing and may not be complete. There maybe some documented evidence that standards are used, but it may not be complete.	The developer organisation(s) responds to reported issues with updates/patches that are usually made available in a reasonably timely fashion. Some support is available, but may be intermittent and without guarantees of continuation. There is evidence of an informal user community that provides answers, for example, via a mailing list or bulletin board. Documentation and source code availability may be sufficient for an experienced user (developer, operations or end-user) to not require extensive support.	Software application demonstrated and tested in a laboratory context. Testing includes testing for error conditions and proof of handling of unknown input.	Security was addressed in the development phase up to and including implementation. Developers have undertaken appropriate security training.	Software is locale aware
Agreements on development responsibilities, the list of developers, a recommended citation, and intellectual property rights statements, affirming the (ed) rights for use, are available, (without upon request), for review. Developer organisation(s) (or developers) may be contacted through a single point to obtain formal statements on restricted rights or to negotiate additional rights.	Future extensibility is designed into the system for a moderate range of use cases. The procedures for extending the software are defined, whether by source code modification or through the insertion of some type of extension functionality (e.g., callback hooks or scripting capabilities). Where source code modification is part of the extension plan, the software is well-structured, has a moderate to high level of cohesion, and has configuration elements clearly separated from logic and display elements.	There is evidence that the software is open, with full structuring into individual components that provide functions or services to outside entities (i.e., user, architecture, internal functions or services documented, but not consistently modular; have been created for generic functions, but modules have not been created for all of the specified functions; code within each module contains many independent logical parts.	The software is easily configurable for different contexts such as, locations of resources (files, directories, URLs) are configurable. All configuration specific information is centralised.	The software is moderately portable. The software can be ported with only relatively small changes necessary to the context or the software itself. Documentation on porting exists and is complete, but requires considerable effort and expertise. Some rudimentary understanding of the underlying software or the target system may be necessary.	The software and software development process complies with some recognised or proprietary standards, but there is incomplete verification of compliance. Compliance to recognised standards has be tested but this may not be for all components. There is documented evidence of standards being used, but not of the verification of compliance.	Support is centralised in a website containing relevant resources, answers to FAQ, other useful information and a community support question & answer area (e.g. bulletin or message/discussion board). There is evidence that the developer organisation(s) occasionally engage with users in the community support area. There are regular scheduled releases of updates/patches that are made available and urgent (serious due to security issues) updates/patches in a timely fashion. There is no opportunity to obtain a support Service Level Agreement (SLA) with the developer(s) or a third party.	Software application demonstrated, tested and validated in a relevant context.	Security was addressed in the development phase up to and including verification and testing.	Content to locale (text, layout, graphics and multimedia, keyboard shortcuts, fonts, locale data and character sets, build process) has been internationalised.
There are multiple licenses or extracted into the software product documents or text that rights and any conditions for use, including commercial and non-commercial use, and the recommended citation. The list of developers is included in the source code of the product, in the documentation, and in the repository of the software open execution. The intellectual property rights statements are expressed in legal language, machine-readable code, and or special statements in language that can be understood by laypersons, such as a pre-written, recognisable license.	There is evidence that the software has been extended/extensible by users outside of the original development group using existing documentation only. There is a clear approach for modifying and extending functions, across one or multiple code bases, with specific documentation and features to allow the building of extensions which are used across a range of scenarios by multiple user groups. There is (or is a library available of non-generated) contact for extensions and not generated documentation on extension is also available.	It is evident that all functions are fully encapsulated into objects or accessible through calls on one or two bases. There is consistent or no handling with meaningful messages and internal and use of generic extensions, in program language, for a larger type checking and compilation. There are or all checking. Functions are available in one calls and each within each module contains few independent logical paths.	A user interface/portal the installer through all the pre-requisites to build, configure and install the software. An uninstaller is also available.	The software is completely portable. The software can be ported to all systems, since it runs as an application layer rather than on the underlying operating system layer. Such software is in form in languages such C#, etc. in Java or Perl. Nevertheless will run on any system in which the appropriate application layer has been installed.	Conformance to open or international recognised standards for the software and software development process. Is tested and documented, and verified through testing of all components. Mostly independent verification is documented through regular testing and certification from an independent group.	The support is (or is provided) in a clearly defined with frequent and timely updates, releases, etc., responding to the needs of the user community, as well as consultation of changes by the community. There is a stated telephone/email helpdesk available as well as a supported website. Discussion groups are active and include regular input from the developer(s) and developer organisation(s). There is evidence that continuity of support is implied. Support was for fixed-fee/limited via a support Service Level Agreement (SLA) with the developer(s) or a third party.	Actual software application tested and validated through successful use of a application in use.	Security was addressed in the development phase up to and including project wide.	Software has been tested with pseudo-localisation.

Colour	Meaning
	Minimum standard
	Expected standard
	Excellent standard

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Maturity Modelling - More Info

Mike Priddy, CESSDA SaW project

Wednesday 15:45

Session 1G - Data Services: Setting up and evaluating

‘Maturity Model For Assessing Data Infrastructures – Cessda As Example’

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What’s Next?

Proposed Phase 2

- more dissemination - understanding and acceptance
- Staging and Production environments
- extend ‘how to’ documentation for developers and admins

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Thank you for listening

Any Questions?

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A pan-European Research Infrastructure for the Social Sciences